A. Bestie

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: brute force, combinatorics, constructive algorithms, implementation, math, number theory |
https://codeforces.com/problemset/problem/1732/A

You are given an array a consisting of n integers $a_1, a_2, \ldots, a_n$. Friends asked you to make the greatest common divisor (GCD) of all numbers in the array equal to 1. In one operation, you can do the following:

- Select an arbitrary index in the array $1 \le i \le n$;
- Make $a_i = \gcd(a_i, i)$, where $\gcd(x, y)$ denotes the GCD of integers x and y. The cost of such an operation is $n - i + 1$.

You need to find the minimum total cost of operations we need to perform so that the GCD of the all array numbers becomes equal to 1.

Input

Each test consists of multiple test cases. The first line contains an integer t ($1 \le t \le 5\,000$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 20$) — the length of the array.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the elements of the array.

Output

For each test case, output a single integer — the minimum total cost of operations that will need to be performed so that the GCD of all numbers in the array becomes equal to 1.

We can show that it's always possible to do so.

Example

Input

```
9
1
1
1
2
2
2 4
3
3 6 9
4
5 10 15 20
5
120 60 80 40 80
6
150 90 180 120 60 30
6
2 4 6 9 12 18
6
30 60 90 120 125 125
```

Output

```
0
1
2
2
1
3
3
0
1
```

Note

In the first test case, the GCD of the entire array is already equal to 1, so there is no need to perform operations.

In the second test case, select $i = 1$. After this operation, $a_1 = \gcd(2, 1) = 1$. The cost of this operation is 1.

In the third test case, you can select $i = 1$, after that the array a will be equal to [1, 4]. The GCD of this array is 1, and the total cost is 2.

In the fourth test case, you can select $i = 2$, after that the array a will be equal to [3, 2, 9]. The GCD of this array is 1, and the total cost is 2.

In the sixth test case, you can select $i = 4$ and $i = 5$, after that the array a will be equal to [120, 60, 80, 4, 5]. The GCD of this array is 1, and the total cost is 3.